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(54) HIGH SPEED BUS VOLTAGE CLAMP FOR POWER TOOL MOTOR DRIVE

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(57) **ABSTRACT**

A power tool comprising, a motor, a DC bus configured to provide operating power to the motor, a source inductor electrically connected to the DC bus, a voltage clamp provided across the DC bus and configured to be selectively activated, and a controller electrically connected to the voltage clamp. The controller is configured to determine that a voltage across the DC bus is greater than a predetermined threshold and activate the voltage clamp in response to the voltage across the DC bus being greater than the predetermined threshold.

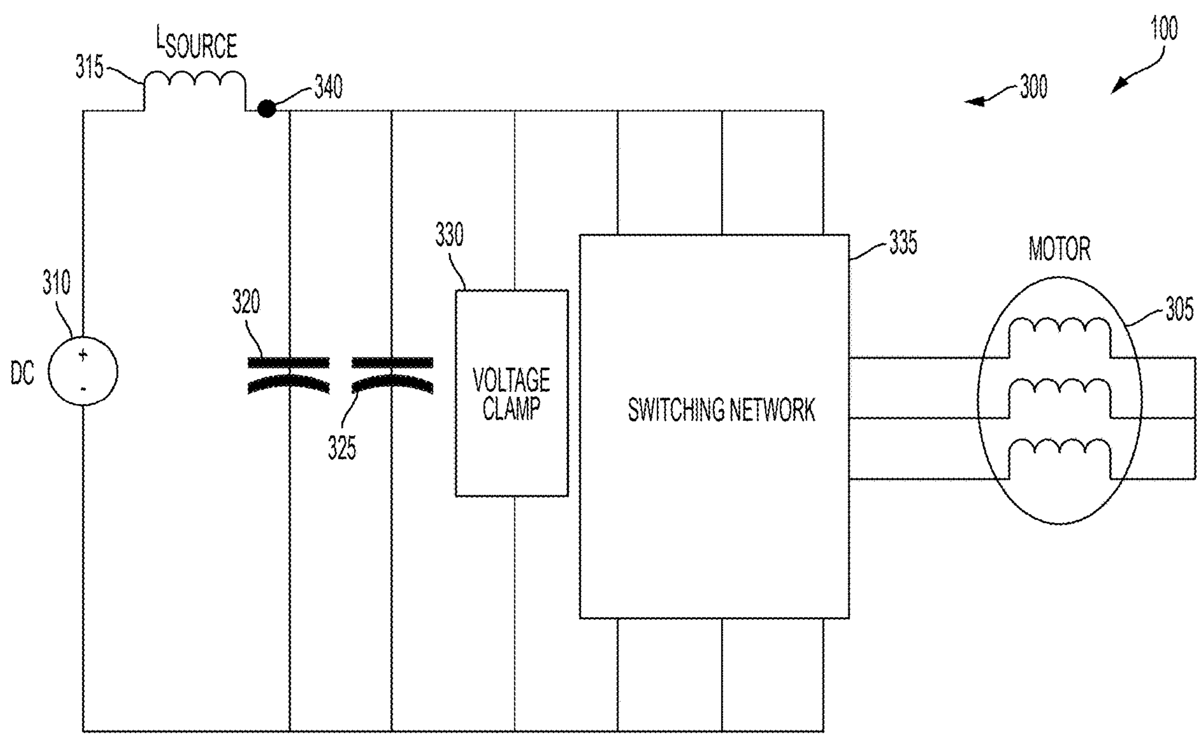


FIG. 1

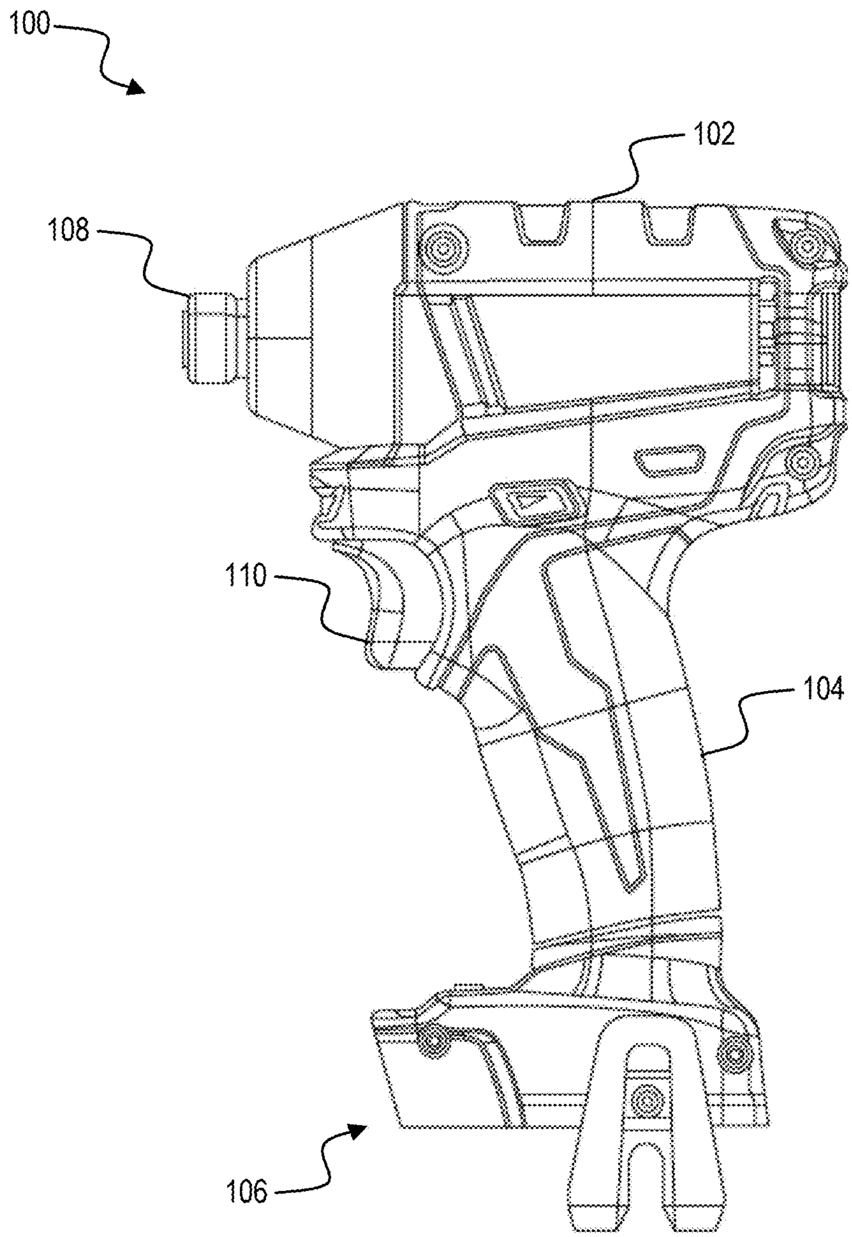
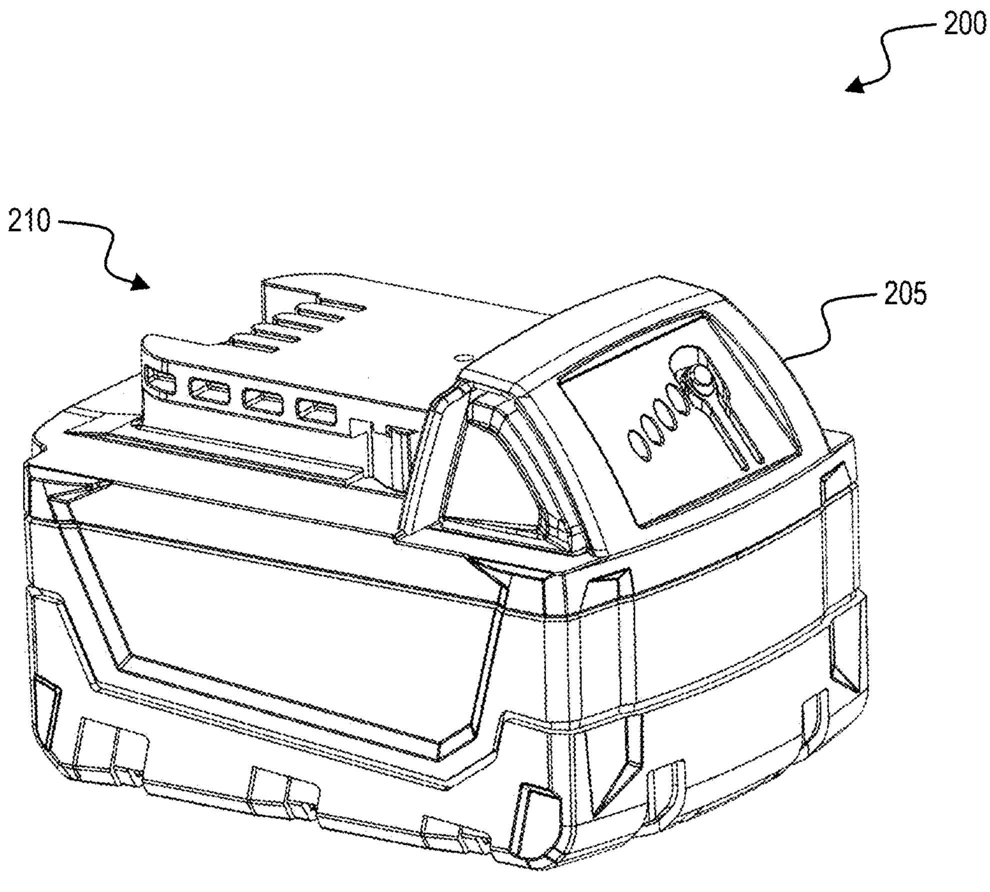


FIG. 2



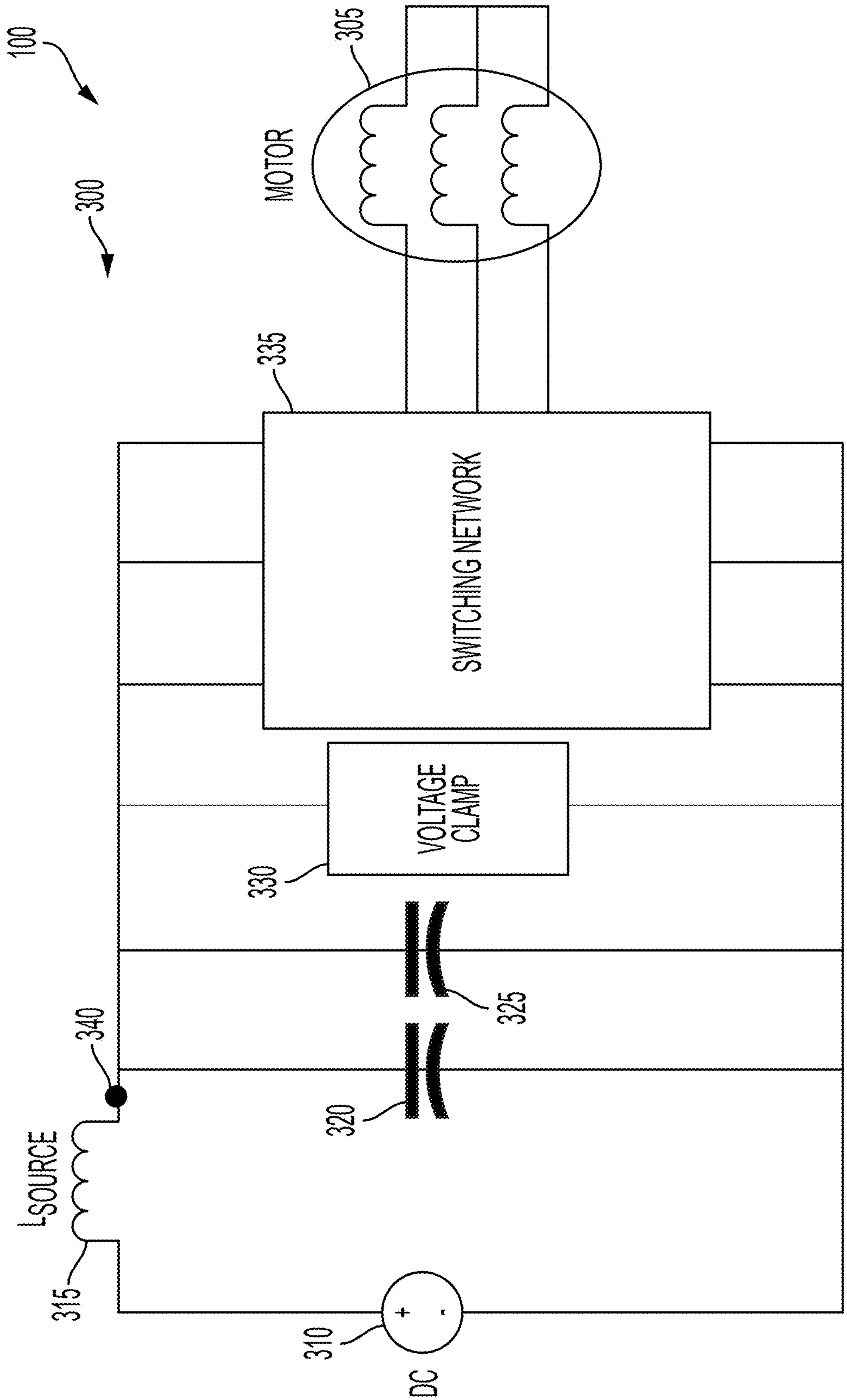

100
300
305
MOTOR
335
SWITCHING NETWORK
VOLTAGE CLAMP
330
325
320
340
L_{SOURCE}
315
310
DC
+
-

FIG. 3A

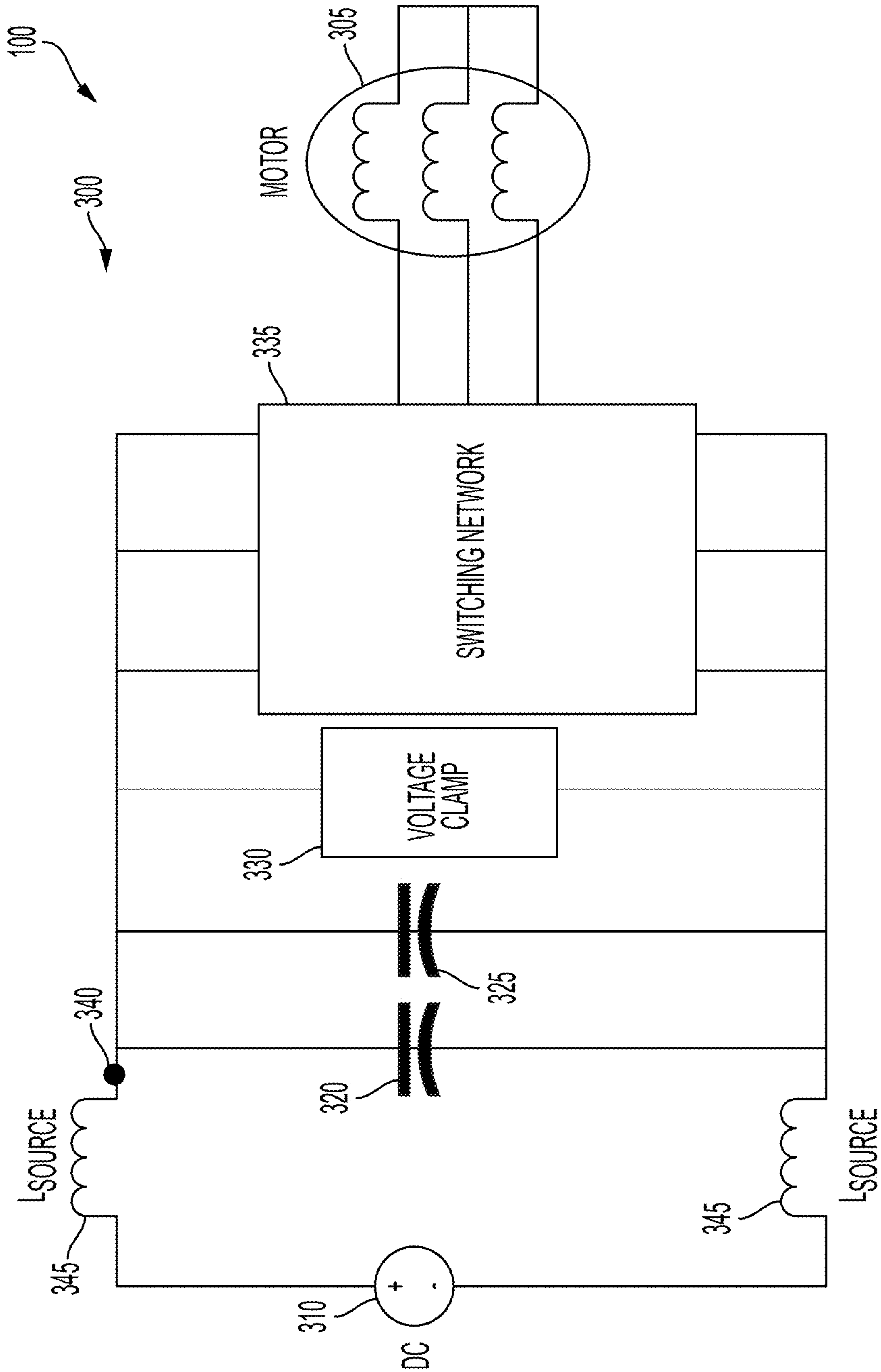

100
300
MOTOR
305
335
SWITCHING NETWORK
VOLTAGE CLAMP
330
325
320
340
L_{SOURCE}
345
L_{SOURCE}
345
310
DC
+
-

FIG. 3B

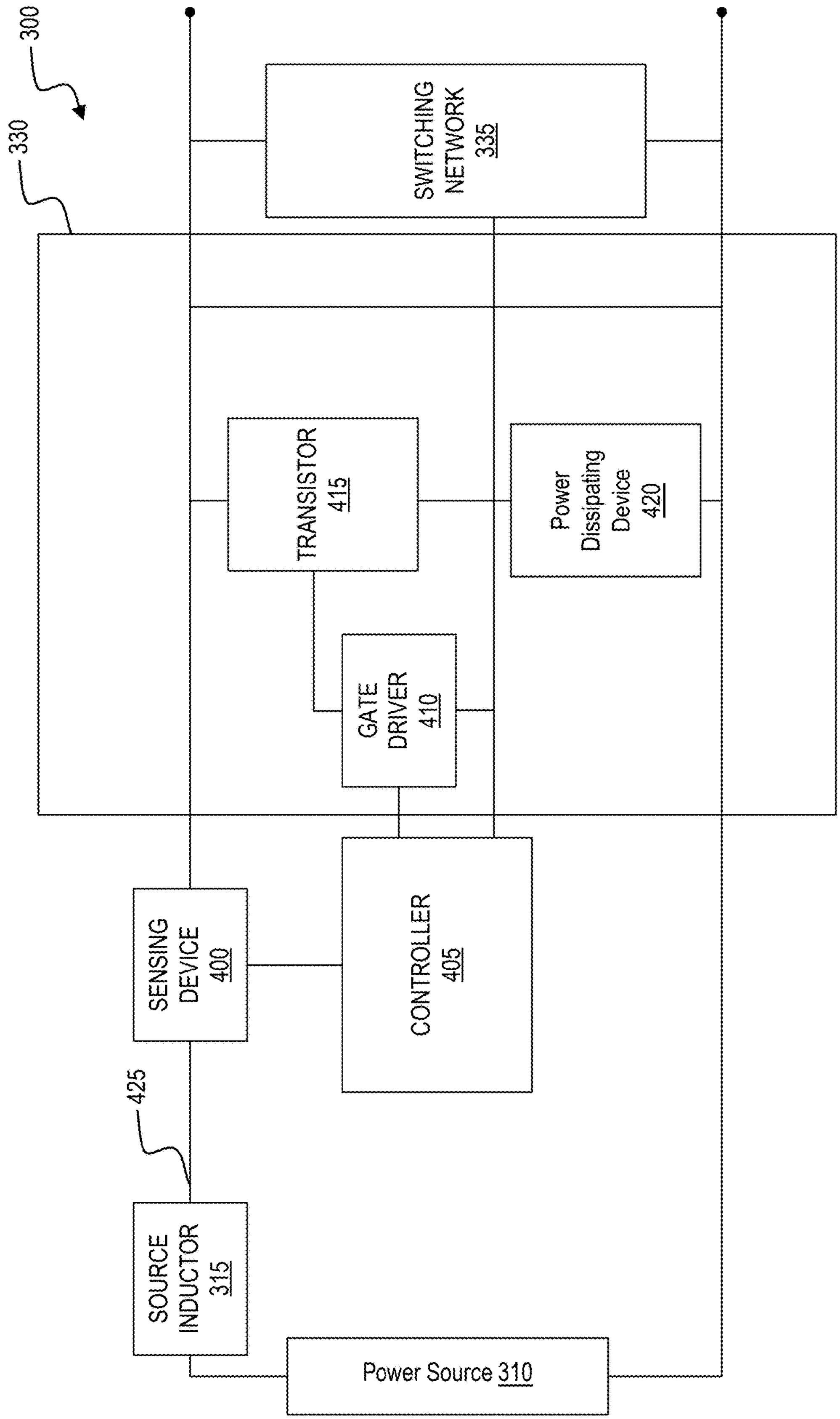

300
330
SWITCHING NETWORK 335
TRANSISTOR 415
Power Dissipating Device 420
GATE DRIVER 410
SENSING DEVICE 400
CONTROLLER 405
425
SOURCE INDUCTOR 315
Power Source 310

FIG. 4

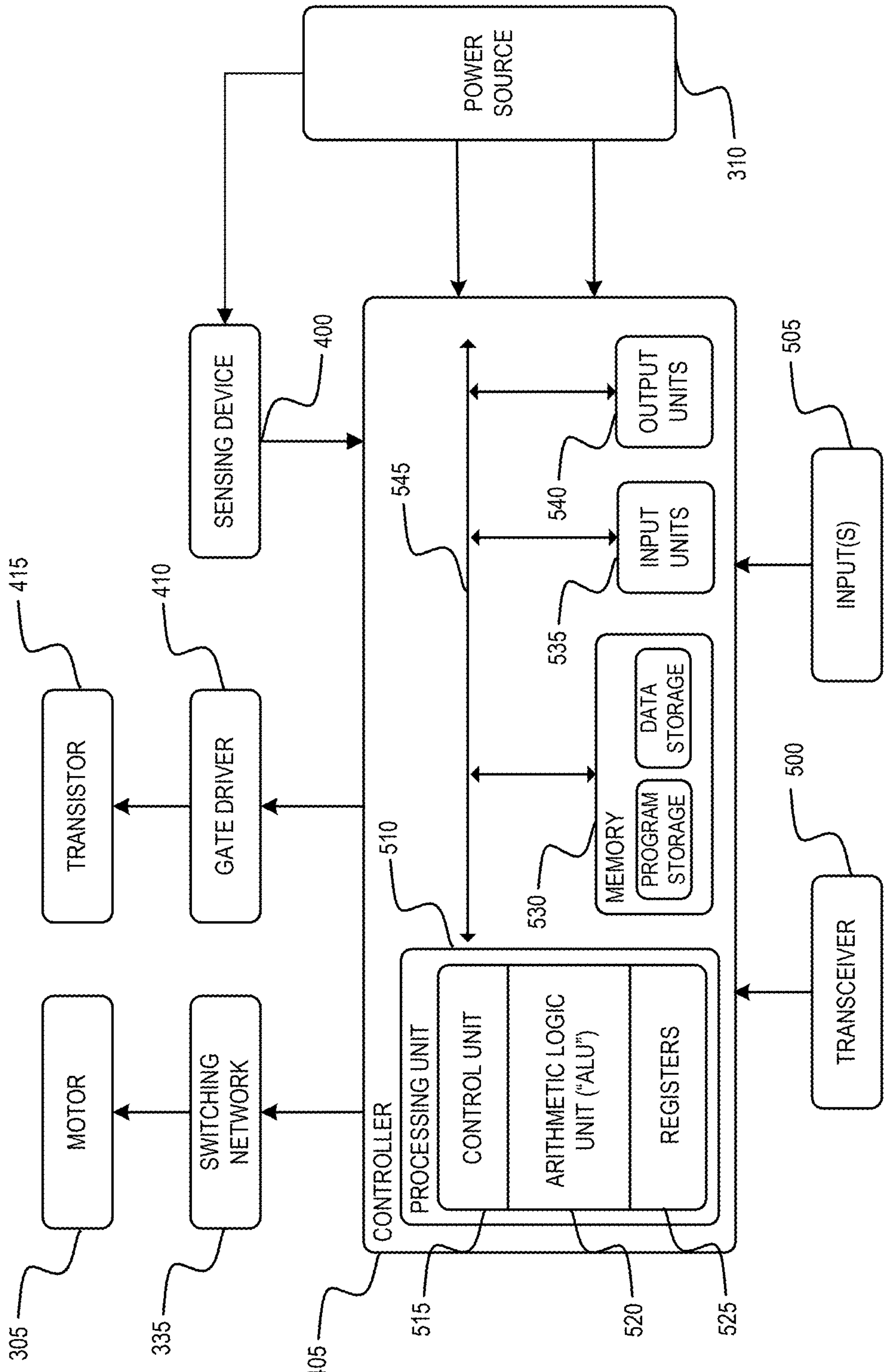

POWER SOURCE
310
SENSING DEVICE
400
CONTROLLER
405
PROCESSING UNIT
510
CONTROL UNIT
515
ARITHMETIC LOGIC UNIT ("ALU")
520
REGISTERS
525
MEMORY
PROGRAM STORAGE
DATA STORAGE
530
INPUT UNITS
535
OUTPUT UNITS
540
545
INPUT(S)
505
TRANSCEIVER
500
GATE DRIVER
410
TRANSISTOR
415
SWITCHING NETWORK
335
MOTOR
305

FIG. 5

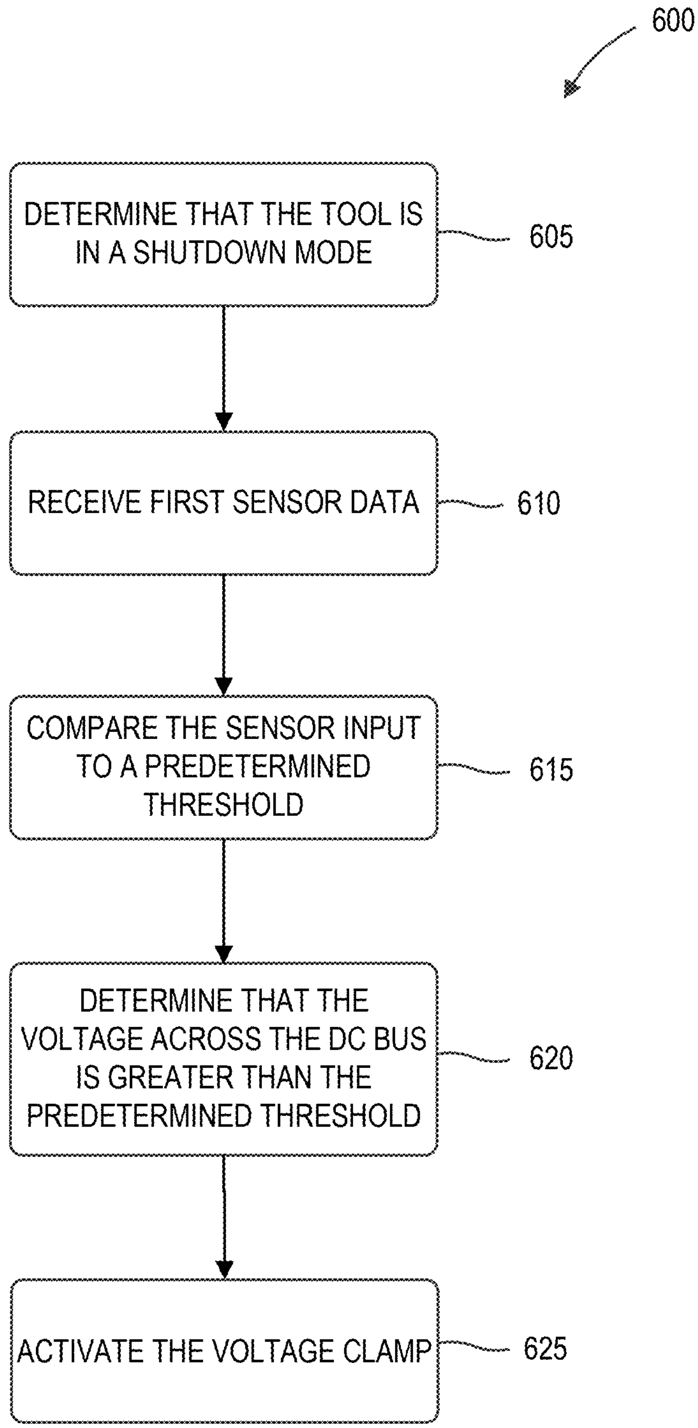

600
DETERMINE THAT THE TOOL IS IN A SHUTDOWN MODE
605
RECEIVE FIRST SENSOR DATA
610
COMPARE THE SENSOR INPUT TO A PREDETERMINED THRESHOLD
615
DETERMINE THAT THE VOLTAGE ACROSS THE DC BUS IS GREATER THAN THE PREDETERMINED THRESHOLD
620
ACTIVATE THE VOLTAGE CLAMP
625

FIG. 6

HIGH SPEED BUS VOLTAGE CLAMP FOR POWER TOOL MOTOR DRIVE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/666,542, filed Jul. 1, 2024, and U.S. Provisional Patent Application No. 63/562,577, filed Mar. 7, 2024, the entire contents of both of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a power tool motor drive during shutdown of a power tool.

BACKGROUND

[0003] Energy becomes trapped in direct-current (DC) power tools when there is a controlled or uncontrolled shutdown, which runs the risk of an overvoltage condition damaging the DC power tool. For example, energy becomes trapped in either the source, load, or both. Traditionally, power tools include DC bus capacitors to absorb the trapped energy.

SUMMARY

[0004] During a controlled or uncontrolled shutdown of a DC power tool, the magnitude of a DC bus voltage needs to be limited so as to not create an overvoltage condition on the DC bus. However, some power tools include multiple capacitors as an energy reservoir to absorb energy that is trapped during a shutdown of the power tool. Capacitors on the DC bus occupy precious space in power tools that are increasingly becoming smaller and more efficient in design. Additionally, capacitors have a parasitic component to them that limits the speed at which they can respond to energy being trapped on the DC bus. Accordingly, there is a need for trapped energy converting solutions that are smaller, lighter, faster, and more efficient for power tools. For example, a transistor is able to facilitate the trapped energy to a power dissipating device that converts the trapped energy into heat, allowing for the DC bus voltage to be at an acceptable level. The single transistor and power dissipating device saves space and provides faster response times compared to the traditional capacitors used as energy reservoirs.

[0005] One embodiment of the present invention discloses a power tool comprising, a motor, a DC bus configured to provide operating power to the motor, a source inductor electrically connected to the DC bus, a voltage clamp provided across the DC bus and configured to be selectively activated, and a controller electrically connected to the voltage clamp. The controller is configured to determine that a voltage across the DC bus is greater than a predetermined threshold and activate the voltage clamp in response to the voltage across the DC bus being greater than the predetermined threshold.

[0006] A further embodiment of the present invention discloses a method for converting trapped energy in a power tool to heat. The method comprises determining, using a controller, that a voltage across a DC bus of the power tool is greater than a predetermined threshold and activating, using the controller, a voltage clamp connected across the DC bus in response to the voltage being greater than the predetermined threshold.

[0007] An even further embodiment of the present invention discloses a power circuit for providing power to a switching network to control a motor. The power circuit comprises a DC bus, a source inductor electrically connected to the DC bus, a voltage clamp, and a controller electrically connected to the DC bus, the source inductor, and the voltage clamp. The controller is configured to determine that a voltage across the DC bus is greater than a predetermined threshold and activate the voltage clamp in response to the voltage across the DC bus being greater than the predetermined threshold.

[0008] Other aspects of the embodiments will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a perspective view of a power tool, according to some embodiments.

[0010] FIG. **2** is a perspective view of a battery pack for powering the power tool of FIG. **1**, according to some embodiments.

[0011] FIG. **3**A is a first simplified block diagram of a power circuit of the power tool of FIG. **1**, according to some embodiments.

[0012] FIG. **3**B is a second simplified block diagram of a power circuit of the power tool of FIG. **1**, according to some embodiments.

[0013] FIG. **4** is a simplified block diagram of a voltage clamp of the power circuit of FIG. **3**, according to some embodiments.

[0014] FIG. **5** is a simplified block diagram of a controller of the power tool of FIG. **1**, according to some embodiments.

[0015] FIG. **6** is a flowchart illustrating a method of voltage clamping of a power tool, according to some embodiments.

[0016] Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein is for the purpose of description and should not be regarded as limited. The use of "including," "comprising" or "having" and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. The terms "mounted," "connected" and "coupled" are used broadly and encompass both direct and indirect mounting, connecting and coupling. Further, "connected" and "coupled" are not restricted to physical or mechanical connections or couplings, and can include electrical connections or couplings, whether direct or indirect. Additionally, unless otherwise noted, terms of approximation, such as "about," approximately," and "substantially," at least when used with numerical values, may refer to within 1%, 2.5%, 5%, or 10% of the noted value.

DETAILED DESCRIPTION

[0017] FIG. **1** illustrates an embodiment of a power tool **100** including a brushless direct current ("BLDC") motor. The power tool **100** is, for example, an impact driver including an upper main body **102**, a handle **104**, a battery pack receiving portion **106**, an output drive device or mechanism **108**, and a trigger **110**. The power tool **100** further includes a motor, such as motor **305** (FIG. **3**) within the main body **102** of the housing and having a rotor and a stator. The rotor is coupled to a motor shaft arranged to produce an output outside of the housing via the output drive device or mechanism **108**. The housing of the power tool **100** (e.g., the main body **102** and the handle **104**) are composed of a durable and light-weight plastic material. The drive device **108** is composed of a metal (e.g., steel) output spindle. The battery pack receiving portion **106** is configured to receive and couple to a battery pack, such as a battery pack **200** (FIG. **2**) that provides power to the power tool **100**. The battery pack receiving portion **106** includes a connecting structure to engage a mechanism that secures the battery pack and a terminal block to electrically connect the battery pack **200** to the power tool **100**. In some embodiments, the power tool **100** may be an alternating current (AC) powered power tool that includes a rectifier to provide a DC voltage to the motor **305**.

[0018] FIG. **1** illustrates an impact wrench, however, the power tool **100** may include drills, circular saws, jig saws, band saws, reciprocating saws, screw drivers, angle grinders, straight grinders, hammers, multi-tools, impact wrenches, rotary hammers, impact drivers, angle drills, powered ratchets, powered torque wrenches, hydraulic pulse tools, hydraulic tensioning tools, lock bolt installation tools, reaction arm tools, riveting tools, nailers, staplers, TC bolt guns, and the like.

[0019] FIG. **2** illustrates a battery pack **200**, according to some embodiments. The battery pack **200** is a power tool battery pack that is generally used to power a power tool, such as the power tool **100**. The battery pack **200** includes a housing **205** and an interface portion **210** for connecting the battery pack **200** to a device (e.g., the power tool **100**). In some embodiments, the battery pack **200** includes lithium ion battery cells. In other embodiments, the battery pack **200** may be of a different chemistry, for example, nickel-cadmium, nickel-metal hydride, and the like. In the illustrated embodiment, the battery pack **200** is an 18 volt battery pack. In other embodiments, the output voltage level of the battery pack **200** may be different. For example, the battery pack **200** can be a 4 volt battery pack, 28 volt battery pack, 36 volt battery pack, 72 volt battery pack or another voltage (voltage here may refer to nominal voltage). The battery pack **200** may also have various capacities (e.g., 3, 4, 5, 6, 8, or 12 ampere-hours).

[0020] The battery pack **200** also includes terminals to connect to the power tool **100**. The terminals for the battery pack **200** includes a positive and a negative terminal to provide power to and from the battery pack **200**. In some embodiments, the battery pack **200** also includes data terminals to communicate with the power tool **100**. For example, the battery pack **200** may include a microcontroller to monitor one or more characteristics of the battery pack **200** and the data terminals may communicate with the power tool **100** regarding the monitored characteristics.

[0021] FIG. **3**A is a first simplified block diagram of the power tool **100** including a power circuit **300** of the power tool **100**, according to some embodiments. The circuit **300** includes a motor **305**, a power source **310**, a source inductor **315**, a first capacitor **320**, a second capacitor **325**, a voltage clamp **330**, a switching network **335**, and a DC bus **340**. In some embodiments, the motor **305** is a brushless DC (BLDC) motor. In some embodiments, the power source **310** is a battery pack, such as the battery pack **200**. The power source **310** provides direct current (DC) operating voltage to the DC bus **340**, which is then provided to the motor **305** (e.g., motor coils) through the switching network **335**. Alternatively, in some embodiments, the power source **310** is an AC power source and a rectifier is provided that turns AC power into DC power that is provided to the DC bus **340**. The first capacitor **320** and the second capacitor **325** (collectively referred to as "the capacitors **320**, **325**") are both coupled in parallel with each other and in parallel with the voltage clamp **330** and the switching network **335**. The source inductor **315** is coupled in series with the power source **310**. The source inductor **315** is coupled on one end to a positive side of the power source **310**. The source inductor **315** is coupled on another end to the capacitors **320**, **325**, the voltage clamp **330**, and the switching network **335**. The source inductor **315** may have a first inductance value. The voltage clamp **330** is provided in parallel with the power source **310**, the source inductor **315**, the capacitors **320**, **325**, and the switching network **335**. The switching network is provided in parallel with the power source **310**, the source inductor **315**, the capacitors **320**, **325**, the voltage clamp **330**, and the motor **305**. The voltage clamp **330** and the switching network **335** will be described below.

[0022] FIG. **3**B is a second first simplified block diagram of the power tool **100** including a power circuit **300** of the power tool **100**, according to some embodiments. The circuit **300** includes a motor **305**, a power source **310**, a first capacitor **320**, a second capacitor **325**, a voltage clamp **330**, a switching network **335**, a DC bus **340**, and two source inductors **345**. In some embodiments, the motor **305** is a brushless DC (BLDC) motor. In some embodiments, the power source **310** is a battery pack, such as the battery pack **200**. The power source **310** provides direct current (DC) operating voltage to the DC bus **340**, which is then provided to the motor **305** (e.g., motor coils) through the switching network **335**. Alternatively, in some embodiments, the power source **310** is an AC power source and a rectifier is provided that turns AC power into DC power that is provided to the DC bus **340**. The first capacitor **320** and the second capacitor **325** (collectively referred to as "the capacitors **320**, **325**") are both coupled in parallel with each other and in parallel with the voltage clamp **330** and the switching network **335**. A first source inductor **345** is coupled in series with the power source **310** at a positive end of the power source **310** and a second source inductor **345** is coupled in series with the power source **310** at a negative end of the power source **310**. The first source inductor **345** is coupled on another end to the capacitors **320**, **325**, the voltage clamp **330**, and the switching network **335**. The second source inductor **345** is also coupled on another end to the capacitors **320**, **325**, the voltage clamp **330**, and the switching network **335**. The two source inductors **345** may both be the first inductance value, or they may both be a second inductance value that is equal to one half of the first inductance value (e.g., such that multiplying the second inductance value by two equals the first inductance value). The voltage clamp **330** is provided in parallel with the power source **310**, the two source inductors **345**, the capacitors **320**, **325**, and the switching network **335**. The switching network is provided in parallel with the power source **310**, the two source inductors **345**, the capacitors **320**, **325**, the voltage clamp

330, and the motor 305. The voltage clamp 330 and the switching network 335 will be described below.

[0023] FIG. 4 is a simplified block diagram of the voltage clamp 330 (e.g., voltage clamp circuit) of the power circuit 300, according to some embodiments. The power tool 100 further includes a sensing device 400 and a controller 405. The voltage clamp 330 includes a gate driver 410, a transistor 415, and a power dissipating device 420. The power dissipating device may be a resistor, a transient voltage suppressor (TVS) diode, a metal-oxide varistor (MOV), and the like. The sensing device 400 is connected to the DC bus 340 and is configured to detect a DC bus voltage across the DC bus 340. In some embodiments, the sensing device 400 provides first sensor data related to the DC bus voltage to the controller 405. Based upon input from the sensing device 400, the controller 405 may output a signal to the gate driver 410. In some embodiments, the gate driver 410 provides pulse width modulated (PWM) signals to the transistor 415 to switch the transistor 415 at a particular frequency with a particular duty cycle. Alternatively, in some embodiments, the gate driver 410 may send an ON signal to the transistor 415 so that it remains ON (e.g., active). When the transistor 415 is ON, the power dissipating device 420 is connected across the DC bus 340 and converts the trapped energy into heat. For example, the current flowing through the power dissipating device 420 generates heat, which may be dissipated by the power tool, for example, using a heat sink. Though shown connected to a negative side of the power source 310, the power dissipating device 420 may be connected to a positive side of the power source 310. For example, the power dissipating device 420 may swap positions with the transistor 415 such that the power dissipating device 420 is connected to a positive side of the DC bus 340 and the transistor 415 is connected to a negative side of the DC bus 340.

[0024] In some embodiments, the transistor 415 is a wide bandgap semiconductor field effect transistor (FET). Wide bandgap semiconductor FETs are made from, for example, Gallium Nitrite (GaN), Silicon Carbide (SiC), or the like, and have a bandgaps in the range of, for example, about 3-4 electronvolts (eV). Wide bandgap semiconductors exhibit several properties that provide advantages compared to traditional field-effect transistors (FETs) (e.g., MOSFETS). Particularly, wide bandgap semiconductors can be operated at very high frequencies, for example, at 100 kHz, 200 kHz, 400 kHz, and more while losing less energy as heat than MOSFETs operating at lower frequencies, for example, 25 kHz, 50 kHz, and the like. Wide bandgap semiconductors have increased response times when turned ON, compared to traditional FETs.

[0025] Because the wide bandgap semiconductors can be operated at very high frequencies, the transistor 415 provides higher resolution signals at the outputs. Energy storage capacity can be reduced for the same total output energy rating caused by the increased frequency during operation of the wide bandgap semiconductors compared to MOSFETs. Additionally, smaller heat sinks and fans can be used because the wide bandgap semiconductors operate more efficiently than MOSFETs and therefore produce less heat during operation. Accordingly, the size and weight of the components within the power tool 100 can be reduced and efficiency can be improved of the power circuit 300 by replacing MOSFETs with wide bandgap semiconductor devices within the voltage clamp 330.

[0026] The switching network 335 controls the motor 305 based on control signals from a motor controller, such as controller 405. The switching network 335 includes a plurality of electronic switches (e.g., FETs, bipolar transistors, and the like) connected together to form a network that controls the activation of the motor 305 using a pulse-width modulated (PWM) signal. For instance, the switching network 335 may include a six-FET inverter bridge that receives PWM signals from the controller 405 to drive the motor 305. Generally, when the trigger 110 is depressed, electrical current is supplied from the power source 310 to the motor 305 via the switching network 335. When the trigger 110 is not depressed, electrical current is not supplied from the power source 310 to the motor 305.

[0027] The controller 405 for the power circuit 300 is illustrated in FIG. 5. In some embodiments, the controller 405 is the controller for the whole power tool 100 (e.g., there is only a single controller in the power tool 100). Alternatively, in some embodiments, the controller 405 is specific to the power circuit 300, and in particular to the voltage clamp 330 and/or the switching network 335. The controller 405 may electrically and/or communicatively connected to a variety of modules or components of the power tool 100. For example, the illustrated controller 405 is connected to a transceiver 500, input(s) 505 (e.g., user interface, trigger switch, sensors, etc.), the power source 310, the sensing device 400, the gate driver 410, and the switching network 335.

[0028] The controller 405 includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller 405, the power tool 100, and/or the power circuit 300. For example, the controller 405 includes, among other things, a processing unit 510 (e.g., a microprocessor, an electronic processor, an electronic controller, a microcontroller, or another suitable programmable device), a memory 530, input units 535, and output units 540. The processing unit 510 includes, among other things, a control unit 515, an arithmetic logic unit ("ALU") 520, and a plurality of registers 525 (shown as a group of registers in FIG. 5), and is implemented using a known computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit 510, the memory 530, the input units 535, and the output units 540, as well as the various modules connected to the controller 405 are connected by one or more control and/or data buses (e.g., common bus 545). The control and/or data buses are shown generally in FIG. 5 for illustrative purposes. The use of one or more control and/or data buses for the interconnection between and communication among the various modules and components would be known to a person skilled in the art in view of the embodiments described herein.

[0029] The memory 530 is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit 510 is connected to the memory 530 and executes software instruction that are capable of being stored in a RAM of the memory 530 (e.g., during execution), a ROM of the memory

530 (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the power tool **100** can be stored in the memory **530** of the controller **405**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **405** is configured to retrieve from the memory **530** and execute, among other things, instructions related to the control processes and methods described herein. In other embodiments, the controller **405** includes additional, fewer, or different components.

[0030] The transceiver **500** allows for wired or wireless communication between the controller **405** and other devices, for example, the power tool battery pack **200**, heavy duty power tools, heavy duty battery packs, an external device (e.g., mobile phone), and the like. In some embodiments, the transceiver **500** may include both a transmitter and a receiver. In other embodiments, the transceiver **500** may include a single device for transmitting and receiving. In some embodiments, the transceiver **500** receives a threshold DC bus voltage magnitude set by a user via an external device that the controller **405** compares to the DC bus voltage to determine whether to turn on the transistor **415**.

[0031] In some embodiments, the input(s) **505** include a user interface (e.g., display, actuators, etc.) and sensors (e.g., motor current sensors, motor voltage sensors, motor position sensors, temperature sensors, torque sensors, trigger pull sensors, etc.). For example, a trigger pull sensor senses the amount that the trigger **110** is pulled and the force with which the trigger is pulled. In some embodiments, the controller **405** determines a shutdown condition of the power tool **100** based on the trigger pull sensor. For example, the controller **405** may determine that the power tool **100** is no longer being operated based on a lack of input from the trigger pull sensor and/or a release of the trigger pull sensor for a predetermined amount of time.

[0032] The motor **305** may be energized based on a state of the trigger **110**. Generally, when the trigger **110** is activated, the motor **305** is energized, and when the trigger **110** is deactivated, the motor **305** is de-energized. In the illustrated embodiment, the trigger **110** is biased (e.g., with a biasing member such as a spring) such that the trigger **110** moves in a second direction away from the handle of the power tool **100** when the trigger **110** is released by the user.

[0033] The switching network **335** enables the controller **405** to control the operation of the motor **305**. The switching network **335** includes a plurality of electronic switches (e.g., FETs, bipolar transistors, and the like) connected together to form a network that controls the activation of the motor **305** using a pulse-width modulated (PWM) signal. For instance, the switching network **335** may include a six-FET bridge that receives PWM signals from the controller **405** to drive the motor **305**.

[0034] The gate driver **410** activates the transistor **415** based on input from the controller **405**. In some embodiments, the gate driver **410** is a power amplifier that receives a low-power input from the power source **310**, via the controller **405** and produces a high-current drive input for the transistor **415**.

[0035] Power tools, such as the power tool **100**, maintain a DC bus voltage during normal operation. However, in the event of a shutdown energy may become trapped in either the source, the load, or both, creating inductance that can cause an overvoltage condition on a DC bus, such as the DC bus **340**. This trapped energy can be removed as heat as further described below.

[0036] FIG. **6** illustrates a method **600** for voltage clamping of the power tool **100** according to some embodiments. Voltage clamping of the power tool **100** turns trapped energy in heat, thus mitigating any potential over voltage that could arise during a shutdown of the power tool **100**. Although the illustrated method **600** includes specific steps, not all of the steps need to be performed or need to be performed in the order presented. In some embodiments, the method **600** is executed by the controller **405**. The values of various thresholds or ranges described with respect to method **600** are provided as non-limiting examples and other values for such parameters are possible and may vary based on the type or condition of the power tool **100**, other factors, or a combination thereof.

[0037] The method **600** includes determining, using the controller **405**, that the power tool **100** is in a shutdown mode (step **605**). In some embodiments, the shutdown is a controlled shutdown of the power tool **100** (e.g., turning the power tool **100** OFF). For example, a controlled shutdown of the power tool **100** may be when a user releases the trigger **110** of the power tool **100**, thus, deactivating the trigger **110**. The controller **405** may determine that the user released the trigger **110** based on an input from the inputs **505** (e.g., trigger pull sensor). In some embodiments, the shutdown may be an uncontrolled shutdown of the power tool **100**. For example, an uncontrolled shutdown of the power tool **100** may be when the controller **405** determines an over-current condition, an over-temperature condition, a kickback condition, or the like of the power tool **100**. The controller **405** determines uncontrolled shutdown of the power tool **100** based on second input data from at least one of the inputs **505** (e.g., voltage sensor, current sensor, temperature sensor, etc.). In some embodiments, when the power tool **100** shuts down, energy may become trapped in the power source **310**, the source inductor **315**, and/or the motor **305**. When trapped energy is not dealt with, an overvoltage condition on the DC bus **425** may occur.

[0038] In step **610**, the method **600** includes receiving, using the controller **405**, first sensor data. In some embodiments, the controller **405** receives the first sensor data from the sensing device **400**. For example, the sensing device **400** might be a voltage sensor that senses a DC bus voltage across the DC bus **425**. The first sensor data is indicative of the voltage across the DC bus **425**.

[0039] In step **615**, the method **600** includes comparing, using the controller **405**, the sensor input to a predetermined threshold. In some embodiments, the predetermined threshold is set by a user on an external device (e.g., mobile phone) via mobile application that transmits the predetermined threshold to the transceiver **500**. Alternatively, or additionally, in some embodiments, the predetermined threshold is set by a user via a user interface of the power tool **100**. In some embodiments, the predetermined threshold is preprogrammed during manufacturing. In step **620**, the method **600** includes determining, using the controller **405**, that voltage across the DC bus **340** is greater than the predetermined threshold based on the comparison in step **615**. In some embodiments, the controller **405** determines that the DC bus voltage is greater than or equal to the predetermined threshold. In some embodiments, the method **600** includes determining, using the controller **405**, that the voltage across the

DC bus **340** is greater than the predetermined threshold based on a leakage current across a MOSFET or other FET provided in the power circuit **300**. For example, the FETs of the switching network **335** or the voltage clamp **330**.

[0040] In step **625**, the method **600** includes activating, using the controller **405**, the voltage clamp **330**. For example, the controller **405** outputs a first signal to a gate driver **410** to turn ON the transistor **415**. The controller **405** activates the voltage clamp **330** in response to the DC voltage being greater than the predetermined threshold (the comparison in step **615**). The controller **405** selectively activates the voltage clamp **330**. In some embodiments, the controller **405** outputs the first signal to the gate driver **410** to turn ON the transistor **415** in response to activating the voltage clamp **330**. In some embodiments, the controller **405** outputs a voltage (e.g., a pulse or a constant voltage) to the gate driver **410** such that the gate driver **410** turns the transistor **415** ON. In some embodiments, the gate driver **410** outs a PWM signal to the transistor **415** at a first frequency. For example, the first frequency of the PWM signal by be between 100 kHz-400 kHz. When the transistor **415** is turned ON, a power dissipating device, such as power dissipating device **420**, is connected across the DC bus **425** and converts energy that is trapped into heat. When turned ON, the transistor **415** creates a current path through the power dissipating device **420**, which gives the trapped energy (e,g., in the source inductor) a place to go (e.g., turning into heat). The power dissipating device **420** being connected across the DC bus **425** also limits the magnitude of the voltage of the DC bus to a first, acceptable level. For example, in a power tool **100** that operates around 18V, an acceptable level may be ±1V.

[0041] Thus, various embodiments described herein provide for power converter devices having wide bandgap semiconductors.

What is claimed is:

1. A power tool comprising:

a motor;

a DC bus configured to provide operating power to the motor;

a source inductor electrically connected to the DC bus;

a voltage clamp provided across the DC bus and configured to be selectively activated; and

a controller electrically connected to the voltage clamp, the controller configured to:

determine that a voltage across the DC bus is greater than a predetermined threshold, and

activate the voltage clamp in response to the voltage across the DC bus being greater than the predetermined threshold.

2. The power tool of claim **1**, wherein the voltage clamp across the DC bus includes a gate driver, a transistor, and a power dissipating device.

3. The power tool of claim **2**, wherein the transistor is a wide bandgap transistor.

4. The power tool of claim **2**, wherein the controller is further configured to:

output, in response to activating the voltage clamp, a signal to the gate driver to turn the transistor on.

5. The power tool of claim **1**, wherein activating the voltage clamp converts energy trapped in at least one of the source inductor and the motor to heat.

6. The power tool of claim **1** further comprising: a sensor configured to generate sensor data indicative of a voltage across the DC bus, wherein the sensor is electrically connected to the controller.

7. The power tool of claim **6**, wherein the controller is further configured to:

determine that the power tool is in a shutdown mode,

receive the sensor data from the sensor, and

compare the sensor data to the predetermined threshold,

wherein the sensor data is indicative of the voltage across the DC bus.

8. The power tool of claim **7**, wherein the shutdown mode is one of a controlled shutdown mode and an uncontrolled shutdown mode.

9. The power tool of claim **1**, wherein the predetermined threshold is set via a user interface.

10. A method for converting trapped energy in a power tool to heat, the method comprising:

determining, using a controller, that a voltage across a DC bus of the power tool is greater than a predetermined threshold, and

activating, using the controller, a voltage clamp connected across the DC bus in response to the voltage being greater than the predetermined threshold.

11. The method of claim **10**, wherein the voltage clamp across the DC bus includes a gate driver, a transistor, and a power dissipating device.

12. The method of claim **11**, wherein the transistor is a wide bandgap transistor.

13. The method of claim **11**, wherein the gate driver controls the transistor with a PWM signal.

14. The method of claim **11** further comprising:

outputting, using the controller, in response to activating the voltage clamp, a signal to the gate driver to turn the transistor on.

15. The method of claim **14**, wherein the power dissipating device is connected across the DC bus converting trapped energy into heat and limiting a magnitude of the voltage across the DC bus to a first level.

16. A power circuit for providing power to a switching network to control a motor, the power circuit comprising:

a DC bus;

a source inductor electrically connected to the DC bus;

a voltage clamp; and

a controller electrically connected to the DC bus, the source inductor, and the voltage clamp, the controller configured to:

determine that a voltage across the DC bus is greater than a predetermined threshold, and

activate the voltage clamp in response to the voltage across the DC bus being greater than the predetermined threshold.

17. The power circuit of claim **16**, wherein the voltage clamp across the DC bus includes a gate driver, a transistor, and a power dissipating device.

18. The power circuit of claim **17**, wherein the controller is further configured to:

output, in response to activating the voltage clamp, a signal to the gate driver to turn the transistor on,

wherein activating the voltage clamp converts energy trapped in at least one of the source inductor and the motor to heat.

19. The power circuit of claim **17**, wherein the voltage clamp includes a sensor configured to generate sensor data indicative of a voltage across the DC bus, wherein the sensor is electrically connected to the controller.

20. The power circuit of claim **19**, wherein the controller is further configured to:

determine that a power tool is in a shutdown mode,

receive the sensor data from the sensor, and

compare the sensor data to the predetermined threshold, wherein the sensor data is indicative of the voltage across the DC bus.

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